Interpretable Gene Programs or Foundation Models?

Comparing Sparse Dictionary Learning and scGPT
for Single-Cell Classification

Jonathan Ma Sijia Zhu Jiatong Gao Yunwei Chai

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Instructor: Dr. Sichen Yang
Draft report scaffold

Abstract

TBD: Write a concise abstract stating the question, controlled comparison, dataset, main quantitative result, interpretability finding, and conclusion. Compare processed expression, sparse dictionary codes, and frozen scGPT embeddings using the same logistic-regression model family. Insert final results only after model selection is frozen and the final evaluation is complete.

1 Introduction

TBD: Motivate the tradeoff between predictive utility and directly inspectable gene programs in single-cell representation learning. State the research question: what does the pretrained scGPT representation add beyond a sparse representation learned from the target training data?
TBD: State the final contributions in two or three concrete points: a controlled classification comparison, an analysis of sparse gene programs, and a reproducible evaluation. Revise these to match the work actually completed.

2 Related Work

TBD: Briefly connect Lasso and sparse coding to dictionary learning (Tibshirani; Olshausen and Field; Mairal et al.), gene-program analysis (Kotliar et al.), scGPT (Cui et al.), and pretraining-scale evaluation (DenAdel et al.). Add verified citations from the literature in jm.tex. Explain how our signed dictionary differs from nonnegative factorization.

3 Data and Shared Evaluation Cohort

3.1 Dataset and provenance

The project uses the Baron human pancreas dataset (GSE84133), obtained through the `scRNAseq` function `BaronPancreasData`. The prepared dataset contains 8,569 cells, 20,125 genes, four donors, and 14 annotated cell types.
TBD: Cite the original dataset and software; summarize count validation, metadata preservation, and any relevant limitations of the annotations.

3.2 Exclusions and fixed splits

Classes with fewer than 30 cells are excluded from the primary comparison, removing 63 cells from the T-cell, Schwann, epsilon, and mast classes. The remaining 8,506 cells span ten classes. A cell-type-stratified random split with seed 42 assigns 5,954 cells to training, 1,277 to validation, and 1,275 to test, approximating

a 70/15/15 allocation. Donors can occur in multiple splits; this evaluates new cells from the same donor pool, not unseen-donor transfer. No additional cells were excluded by the scGPT panel check.
TBD: Confirm these counts against the final run manifest. Explain the practical rare-class cutoff and small remaining evaluation classes. Keep the split's scientific limitation in the main report, not only in an appendix.

4 Methods and Theoretical Basis

4.1 Classical expression preprocessing

The current baseline normalizes each cell to 10,000 total counts across the original gene set, applies $\log(1+x)$, selects 2,000 genes by training-set log-expression variance, and uses training-fitted gene-wise centering and scaling. Identical fitted transforms are applied to validation and test cells.
TBD: Confirm or revise this provisional recipe after coordination with the classifier teammate. Specify that variance ranking is not dispersion-adjusted HVG selection. Explain that scGPT has separate input processing.

4.2 Sparse dictionary learning

Let $X \in \mathbb{R}^{n\times p}$ contain processed expression, with cells in rows. We represent it by ZD, where $Z \in \mathbb{R}^{n\times K}$ contains cell activities and $D \in \mathbb{R}^{K\times p}$ contains gene-program atoms:

$$\min_{Z,D} \frac{1}{2}\|X - ZD\|_F^2 + \lambda\|Z\|_1, \qquad \|d_k\|_2 \le 1 \quad (k = 1, \dots, K). \tag{1}$$

The atom constraint prevents reducing the sparsity penalty by rescaling the factors. With D fixed, each cell's code solves a convex Lasso problem; learning both factors is nonconvex. Validation and test codes must use the learned training dictionary without updating its atoms.
TBD: Describe the final mini-batch solver, signed coefficients, stopping rule, initialization, and selected K and λ. Briefly explain the reconstruction–sparsity tradeoff and limits of convergence claims. Keep this core argument here; put detailed updates and derivations in the appendix.

4.3 Frozen scGPT embeddings

The supplied comparison checkpoint is `sctab_seed0_n1600000_scgpt`: a small custom scGPT model pretrained on 1.6 million cells with sampling seed 0. It is not the original published large scGPT checkpoint. Its frozen representation uses 128-dimensional, L2-normalized CLS embeddings. The Baron input matches 1,879 of the fixed 2,000 panel genes; missing panel genes are zero-filled by the extractor.
TBD: Cite scGPT, summarize the checkpoint choice made without Baron classification scores, and describe binning and extraction settings. Verify against the supplied metadata. Pretraining overlap remains unaudited unless a subsequent audit resolves it; do not claim independent pretraining/test data.

4.4 Common logistic classifier

TBD: Describe three separately fitted L2-regularized multinomial logistic classifiers: processed expression directly, dictionary codes, and scGPT embeddings. Specify the common tuning procedure, class ordering, any training-fitted representation scaling, and optimization settings. Do not concatenate representations unless explicitly introducing a separate experiment.

5 Experimental Design

TBD: Describe validation-only selection and the agreed search budget. Record revisions to the provisional dictionary grid after the classifier teammate's initial validation results. The starter grid is $K \in \{16, 32\}$

and $\lambda \in \{1, 5\}$, seed 42, with a ten-epoch ceiling; this is not yet the final model-selection result. State how classifier regularization is tuned consistently across representations.
TBD: Specify Macro-F1 as primary, plus balanced accuracy and multiclass log loss. Explain cell-ID alignment, identical evaluation cells, and the rule that final test results are inspected only after choices are frozen. If reporting uncertainty or repeated runs, define the procedure and what variability it captures; do not assume cells from the same donor are independent.

6 Results

6.1 Classification performance

TBD: Lead with the strongest quantitative comparison and answer the research question. Keep validation selection separate from final held-out performance. Do not fill this table with reconstruction metrics or smoke-test outputs.

Table 1: Final held-out classification results (not yet evaluated). Higher Macro-F1 and balanced accuracy are better; lower log loss is better.

Representation	Macro-F1	Balanced accuracy	Log loss
Processed expression	TBD	TBD	TBD
Sparse dictionary codes	TBD	TBD	TBD
Frozen scGPT embeddings	TBD	TBD	TBD

6.2 Predictive utility and sparsity

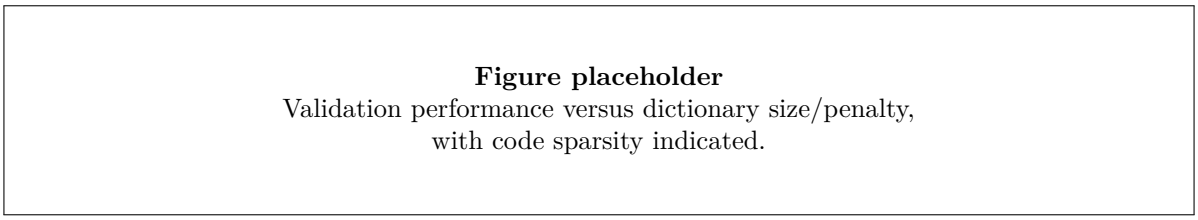


Figure 1: TBD: show the principal tradeoff supporting the selected dictionary. Specify the evaluation split and uncertainty, if any.

TBD: Discuss whether compression and sparse activity preserve useful cell-type information. Report reconstruction as a diagnostic rather than evidence of classification success. Summarize convergence and stability only where tested.

6.3 Gene-program interpretation

TBD: Discuss a small number of informative programs with biological support where available. Account for atom sign/permutation ambiguity. Avoid equating associations with causal mechanisms or claiming every atom is a unique pathway.

7 Discussion and Limitations

TBD: Explain the performance–interpretability tradeoff supported by the final results, including whether scGPT adds linearly accessible predictive information. Discuss why the expression baseline matters. Avoid conclusions based solely on model size or reconstruction.

Figure placeholder
Selected atoms: high-weight genes and
cell-type activation patterns.

Figure 2: TBD: strongest interpretable gene-program evidence. Distinguish positive and negative weights and describe the selection criterion for displayed atoms.

TBD: Keep major limits in the main text: shared donors across splits, excluded rare classes, small per-class evaluation samples, missing panel genes, custom-checkpoint scope, unaudited pretraining overlap, and validation-search limitations. Add computational or stability limitations observed in final runs.

8 Conclusion

TBD: Give the direct answer to the research question and the principal evidence in a short paragraph. State what remains unresolved. Do not introduce a new experiment or essential qualification only in the appendix.

References

TBD: Insert the verified, cited bibliography. Start from jm.tex, and add the Baron dataset and implementation/software references actually used. Ensure every in-text citation resolves and every claim is supported by the relevant source.

A Supplementary Theory

TBD: Detailed sparse-coding derivation, proximal/alternating updates, atom-norm constraint reasoning, and additional optimization discussion. The main text must retain the objective and key theoretical insights.

B Implementation and Reproducibility

TBD: Commands, package versions, checkpoint/data/manifest hashes, artifact schemas, seeds, batch sizes, precision, and hardware. Record the final code revision and access instructions without including private credentials.

C Supplementary Experiments and Figures

TBD: Complete candidate-grid results, per-class metrics, confusion matrices, additional atoms, convergence diagnostics, and seed sensitivity where available. Only supplementary evidence belongs here; the strongest results remain in the main report. Clearly label analyses that were not performed.

D Team Contributions

TBD: Summarize each member's actual contributions to preprocessing, scGPT, dictionary learning, classification, verification, and writing.